

CMS-PRO

Function-Aware Citation Retrieval and Generation

Anushka Pankaj Joshi

Faculty of Computer Science,
Technische Universität Dresden
anushka.pankaj_joshi@mailbox.tu-dresden.de

Examiner: Prof. Dr.-Ing. Michael Färber

Supervisor: Tobias Schreieder

Abstract

Large language models often struggle to give answers with clear, reliable citations, and even correct references can be misaligned with a claim’s intent. We address *Scientific Question Answering (SciQA)* with a *function-aware Retrieval-Augmented Generation (RAG)* pipeline. We perform standard semantic retrieval followed by *function-conditioned filtering/verification*, then generation of a unified answer with *function-tagged spans*. Conditioning uses the *citation function* (role/purpose: Background, Compares, Extends, Uses, Motivation, Future Work) to align each cited source to the intended rhetorical role. *Support* is evaluated at the function level. A function is counted as supported iff at least one cited abstract sentence directly justifies its tagged span. We report per-function support rates and the fraction of supported functions per answer. On a dataset of 400 question–citation function pairs, our system achieves a Support F1 of 0.849 and a function-classification accuracy of 0.918. A human-annotated sample (50 questions, 100 citation function) confirms these results, suggesting robustness. To support reproducibility, we release a gold evaluation set built with constrained labeling and post-processing using LLM, plus templates for creating similar datasets. Overall, function-aware conditioning improves *citation relevance* and evidence traceability for *SciQA*.

1 Introduction

Large language models (LLMs) have transformed knowledge-intensive tasks such as scientific question answering (SciQA), literature summarization, and scientific-discovery support (Lála et al., 2023; Zheng et al., 2023; Wang et al., 2024). RAG grounds outputs in external documents to reduce hallucinations and improve factuality (Gao et al., 2023, 2024; Shuster et al., 2021), but most systems optimize for answering and bibliographic

correctness while treating retrieved citations as interchangeable. This weakens *trustworthiness* by harming *citation relevance* and *verifiability*. In scientific writing, citations serve distinct rhetorical functions: Background, Uses, Compares, Extends, Motivation, Future Work (Teufel et al., 2006; Cohan et al., 2019; Jurgens et al., 2018). Ignoring these functions can yield answers that are technically referenced yet misaligned with a claim’s intent, reducing reliability for *SciQA*. We therefore frame our goal as SciQA with *function-appropriate* citations, emphasizing relevance and verifiable grounding.

Parallel strands of research address parts of this challenge. Here, *attribution* denotes mapping each generated claim to a concrete, verifiable evidence span with provenance which is distinct from *citation generation* (selecting/formatting references) and from *grounded text generation* (RAG) that aims to support answers with retrieved sources (Gao et al., 2023). Work on attribution and citation evaluation proposes benchmarks and metrics for testing whether generated outputs are supported by retrieved evidence (Gao et al., 2023). Citation-analysis research develops taxonomies, datasets, and models for *citation function/intent*, spanning early rule-based methods (Teufel et al., 2006), **supervised** classifiers (e.g., discourse-role and function datasets (Jurgens et al., 2018)), large neural models such as SciCite (Cohan et al., 2019), and **recent semi-supervised** approaches (Chatrinan et al., 2025). Despite progress, these lines remain largely disconnected. Evidence-grounded generation rarely conditions on rhetorical roles, and citation-function models are not typically integrated into retrieval or verification. Contemporary scientific RAG systems e.g., PaperQA (Lála et al., 2023), BioRAGent (Ateia and Kruschwitz, 2024), and SQuAI (Besrou et al., 2025) show strong passage-level retrieval and QA pipelines, but as far as we are aware do not make retrieval or

selection *function-aware*, leaving a gap between retrieval advances and the nuanced purposes citations serve in scientific communication.

To address this gap, we propose a *function-aware citation retrieval and generation pipeline for SciQA*. The pipeline integrates *citation function* signals into selection and generation as follows: (1) an LLM predicts one or more citation functions for the user query (BACKGROUND, USES, MOTIVATION, EXTENDS, FUTUREWORK, COMPARES); (2) we run **standard semantic retrieval** to obtain top- k candidate papers and rank based on cosine similarity (3) a second LLM performs **function-conditioned filtering/verification** by checking whether each candidate’s abstract *fits the predicted function* and (4) we generate a unified answer with *function-tagged spans* and in-line citations selected from the filtered set. This design choice is practical as large scientific corpora *lack explicit function annotations* at scale, so we infer functions on demand and then verify them against retrieved candidates, which is more efficient than pre-annotating corpora and directly aligns sources to their rhetorical role. By aligning retrieval-time selection and answer-time citation to *citation relevance and verifiable grounding*, our system targets SciQA with function-appropriate evidence rather than generic topical matches.

- We introduce **function-aware citation retrieval and generation** pipeline for SciQA, coupling standard retrieval with function-conditioned filtering/verification.
- We formalize **function-level Support** i.e. a function is supported iff a cited abstract sentence directly justifies its function-tagged span. We report per-function rates and fraction supported per answer.
- We release a LLM derived **gold evaluation set** (constrained labeling + post-processing) and reusable **templates** for creating similar datasets.
- On a dataset of **400 question–citation function pairs**, the system attains **Support F1 = 0.85** and **function-classification accuracy = 0.92**. A human-annotated sample (50 questions, 100 function-claims) confirms these results.

An overview with a worked example appears in Figure 1.

2 Related Work

Retrieval-Augmented Generation. Retrieval-Augmented Generation (RAG) combines a retriever with a generator that *conditions on the retrieved texts* to answer a query (Lewis et al., 2021; Karpukhin et al., 2020; Izacard and Grave, 2021). By grounding the generator in external documents, RAG reduces unsupported content and improves factual correctness and verifiability in knowledge-intensive settings. Scientific RAG systems (e.g., PaperQA (Lála et al., 2023)) typically optimize for strong topical retrieval and answer quality; however, they do not model *citation intent* during selection. In our work, we build on this paradigm but introduce *function-aware* selection after retrieval.

Citation generation with grounded models.

Given grounded evidence, systems must decide *which sources to cite and where*. Approaches differ by when citing is decided: (i) **post-retrieval in-generation**: the model is prompted with retrieved passages and outputs in-line citations tied to those passages;. (ii) **post-generation**: the system first generates an answer, then links claims to sources and assigns citations. (iii) **hybrid/iterative**: generation and retrieval alternate so interim text steers subsequent retrieval (e.g., Iter-RetGen (Shao et al., 2023)). We adopt retrieval-before-generation rather than post-generation retrieval because, on our small, domain-limited corpus, post-generation drafts frequently contained claims not covered by the index, preventing reliable grounding.

For evaluating the *quality of generated citations* (accuracy of author/title and linkage to supporting evidence), benchmarks and frameworks such as ALCE (Gao et al., 2023) and RAGE (Penzkofer and Baumann, 2024) provide task setups and metrics. Methods like AGREE improve grounding and selection via tuning and are orthogonal to our function-aware filtering (Ye et al., 2024). Finally, some work avoids architectural changes to the Language Model (LM) by *in-context* augmentation, prepending retrieved documents and relying on off-the-shelf retrievers (In-Context RALM (Ram et al., 2023)).

Our pipeline is complementary, we keep standard retrieval but insert a *function-conditioned filtering/verification* step before generation so that cited sources are not only accurate and relevant but also aligned with the intended *citation function*.

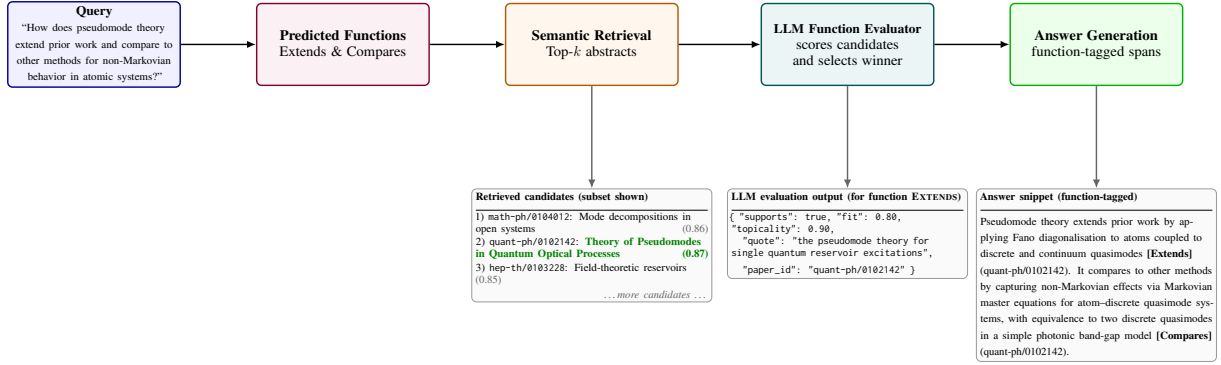


Figure 1: **RAG pipeline for function-aware citation retrieval and generation (SciQA)**. Given a query, the system predicts citation functions, retrieves top- k abstracts, applies function-conditioned filtering/verification, and generates an answer with function-tagged spans and inline citations.

Citation Function Classification. Identifying the rhetorical role of citations in scientific text has a long history. (In prior work this is commonly called *citation intent*. We use *citation function* because our system *generates* text with citations and conditions retrieval/selection on the intended role.) Early rule-based systems leveraged surface cues (Teufel et al., 2006), supervised neural models such as SciCite used local context around the citation (Cohan et al., 2019), and discourse-oriented frameworks examined relations between citing and cited works (Jurgens et al., 2018). More recently, semi-supervised approaches (e.g., GAN-CITE) improve performance with limited labels (Chatrian et al., 2025), while prompt-based methods (CITEPROMPT and follow-ups) show that LLMs can infer intent directly from context (Lahiri et al., 2023; Koloveas et al., 2025). Our use differs in *where* the signal is applied. We predict the *function* from the query/abstract and use it **post-retrieval** for **function-conditioned filtering/verification** and answer generation. This makes citation selection sensitive to the intended role (Background, Uses, Compares, Extends, Motivation, FutureWork) and aims to improve *citation relevance* and *verifiable grounding* in SciQA, rather than only classifying intents in existing text.

Recent work examines two aspects of citation function. First, multiple in-text citations can carry *different* functions within the same sentence. Segment-wise SciBERT variants model this and achieve state-of-the-art results on re-annotated, multi-dataset collections (Jiang et al., 2023; Jiang and Chen, 2023). Second, LLMs can infer functions from local context and provide a lightweight alternative to supervised training (Ikoma and Matsumura, 2023). Our approach differs by using func-

tion signals downstream, after retrieval, for filtering and generation in SciQA.

3 Methodology

Our system follows a modular **four-stage** design that combines citation-function classification, semantic retrieval, function-based filtering, and function-aware answer generation. The goal is to align SciQA with the intended *citation function*, so references are not only factually correct but also role-appropriate.

3.1 Pipeline Overview

The overall pipeline is shown in Figure 2. Given a user question, the system predicts one or more *citation functions*, retrieves top- k candidate abstracts via *semantic retrieval*, verifies which candidates actually support the predicted function, and then generates an answer using only the verified, function-consistent evidence with function-tagged spans and in-line citations.

Stage 1: Citation-Function Classification.

Given a scientific question, an LLM predicts the *citation function* that best characterizes the source needed (we use BACKGROUND, USES, COMPARES, EXTENDS, MOTIVATION, FUTUREWORK). We employ a few-shot prompt to elicit these functions, building on prior work in citation intent/function modeling (Teufel et al., 2006; Cohan et al., 2019; Jurgens et al., 2018) and prompt-based methods (Lahiri et al., 2023; Koloveas et al., 2025). Functions are predicted *before retrieval* and used *after retrieval* to drive filtering and generation. In pilot experiments, a fine-tuned allenai/scibert_scivocab_uncased reached macro-F1 ≈ 0.86 . Further training details for pilot

experiment appear in (Appendix 9, Table 6). We switched to a prompt-based LLM classifier because it performed better in our setup.

Stage 2: Semantic Retrieval. We use `intfloat/e5-small-v2` to embed queries and *title+abstract* for each paper; cosine over ℓ_2 -normalized vectors is used to **rank** candidates. The retriever returns the top- k most similar items ($k=10$) from a corpus built on the *unarXive_2024* Hugging Face release (Besrou and collaborators, 2024), which updates prior unarXive versions (Saier and Färber, 2020; Herrmannova and et al., 2023). Retrieval here is purely semantic and function-agnostic, following standard dense/semantic retrieval practice (Reimers and Gurevych, 2019; Karpukhin et al., 2020; Thakur et al., 2021). The construction of our retrieval corpus and index is detailed in Section 4.

Stage 3: Function-Aware Filtering. After retrieval returns a ranked top- k list, we **filter** candidates with a lightweight symbolic screen and an LLM check. We first extract a compact set of *core entities/terms* from the question (key noun phrases, hyphenations, symbols/abbreviations) via a prompt-based extractor (max 6 items) with a regex fallback. If empty, we fall back to content terms. This anchors judgments to the question’s terminology and reduces off-topic acceptances versus an LLM-only filter. Each candidate (title+abstract) is then evaluated under a strict JSON schema:

- **supports** (bool): whether the quoted text directly justifies the predicted function for this query.
- **fit** (0–1): how well the evidence fulfills the target *function*.
- **topicality** (0–1): closeness to the question/core terms irrespective of function.
- **quote** (≤ 40 words): verbatim substring from title/abstract used as evidence.
- **why** (`off_topic | no_function_evidence | sufficient`): brief rationale for the decision.

We require the **verbatim quote** to appear in the title/abstract; otherwise `supports=false`. This yields span-level, auditable evidence and enables exact-substring checks, while the final answer may paraphrase. We verify up to `max_check` candidates

per function (default 10) in retrieval order, then select at most one *winner* per function by sorting

(`fit+ $\delta_{\text{cue}}$` , `topicality`, `retrieval_cosine`),

with $\delta_{\text{cue}}=0.03$ if the quote contains function-specific cues, else 0. An on-topic arbiter (strict-JSON LLM prompt) rejects winners that do not clearly relate to core entities. If rejected, we consider the next supported candidate.

Stage 4: Function-Aware Citation and Answer Generation. Using only verified, function-consistent evidence, the generator produces a concise answer with function-tagged spans and in-line paper IDs. Unlike prior RAG systems that treat evidence uniformly (Lewis et al., 2021; Izacard and Grave, 2021; Karpukhin et al., 2020; Zheng et al., 2023; Wang et al., 2024; Wadden et al., 2020), our content selection and citation placement are conditioned on the predicted *citation function*. As multi-source answers stress model capacity, we keep generation concise and abstract-grounded (Qian et al., 2024), and our verifier naturally supports future “transparentized” pipelines that distinguish parametric vs. retrieved knowledge in cited spans (Shen et al., 2025). An example for answer generation is shown in Figure 1.

Interfaces. All stage I/O is JSONL and detailed field schemas are in Table 8 (Appendix 9).

3.2 Models and Decoding

Runtime stages (classification, filtering, generation) use `openai/gpt-oss-120b`. Retrieval embeddings use `intfloat/e5-small-v2`. Decoding settings are fixed per stage. Backbone comparisons with alternative LLMs are reported in §6.

3.3 Prompt Design: Rationale

Our prompts are optimized for *SciQA with abstract-level evidence*, and we score *Support at the function level*. The answer must contain a function-tagged span that is directly justified by a verifiable quote from the cited abstract. For *citation function* classification, we state the task and output constraints *before* a few concise positive examples, so the examples reinforce a clear policy rather than determine it. The verifier is designed for auditability. It returns strict JSON (`supports`, `fit`, `topicality`, `quote`, `why`) and must anchor judgments in core entities from the question. Additionally, requiring a short verbatim quote provides span-level provenance and



Figure 2: Methodology overview showing the four core stages of the function-aware citation pipeline. Each stage progressively refines evidence selection and generation according to the predicted citation function.

enables exact-substring checks. To avoid brittleness, we adopt a concrete relaxed rule: a quote may omit the core term if the abstract elsewhere clearly mentions a core entity (exact string or unambiguous synonym) and the quoted sentence is substantively tied to that context. The generation prompt inherits these constraints and conditions on the predicted *citation function*, so content selection and in-line citations remain *function-aligned* rather than merely topically relevant.

This architecture allows each stage to be independently improved or replaced e.g., using more advanced embedding models for retrieval or alternative LLMs for function evaluation while maintaining a unified function-aware framework.

4 Dataset: FACET

FACET (Function-Aware Citation Evaluation and Testbed) is the dataset used throughout this work. It comprises (i) a fixed evidence index of research abstracts used for retrieval and (ii) two evaluation sets: a gold LLM-derived test set and a small human-annotated sample.

4.1 Evidence Index (Retrieval Corpus)

Base corpus. We start from UNARX-IVE_2024 (Saier and Färber, 2020) (full collection $\sim 2\text{M}$ papers) and curate a fixed evidence index of **9,262** research abstracts that all systems query throughout our experiments. Each abstract is lightly normalized to remove LaTeX/math artifacts and boilerplate, then filtered using simple quality thresholds. We retain only items whose cleaned abstract contains at least **600 characters**, **150 words**, and **5 sentences**, to ensure sufficient content density for extractable, verifiable evidence spans and reliable function matching. We saw that very short abstracts frequently yielded no on-topic quote and reduced answerability in pilot runs.

Record schema. Each paper record is stored as `{"paper_id", "title", "authors", "year", "abstract"}`. Title and abstract drive retrieval,

metadata (authors, year, paper_id) supports citation and display.

Embedding construction. For each record we concatenate `<title>+""+<abstract>+""+<authors>+""+<year>+""+<paper_id>` and encode with `intfloat/e5-small-v2`. This captures topical content and bibliographic context, improving queries that reference specific works/authors.

Indexing. Embeddings and metadata are stored in a FAISS index with cosine similarity for top- k search. Each vector retains a pointer to its JSON metadata for downstream access. The evidence index is used only for retrieval (no fine-tuning on the index or labels).

4.2 Evaluation Sets

We construct a held-out test set of **200 questions** with **400 question–citation function pairs** (two per question). For each of 200 sampled papers from the index, we randomly choose two functions from `{BACKGROUND, USES, COMPARES, MOTIVATION, EXTENDS, FUTUREWORK}` and prompt *meta-llama/Llama-3.3-70B-Instruct* to write a single question that jointly implicates both functions, along with an answer that must be grounded strictly in the abstract. Answers contain bracket tags (e.g., "... [Uses]"), which we parse together with explicit insufficiency cues to derive binary supported/unsupported labels per function target. This yields per-function supervision for support and per-question supervision for Citation Function (the set of functions implicated by the question). Marginal function frequencies are approximately balanced by construction. The evidence index is used only for retrieval (no fine-tuning on the index or labels), and all questions are answerable from abstracts alone, avoiding leakage from external full-text sources.

To assess robustness under human judgments, we create a small manually annotated sample comprising **50 papers** with **100 citation functions** (two per paper) in which we *reuse the same LLM-generated questions* but the first author of this report *annotates the function labels and the supported/unsupported tags by hand* directly from the abstracts. This subset serves as an auxiliary check rather than a replacement for the main test set. It verifies that the pipeline behaves sensibly on manually created ground truth. Results for the human-annotated dataset are reported in §6, Tables 4 and 5.

We refer to the retrieval corpus as **FACET (index)** and to the two evaluation sets as **FACET (gold)** and **FACET (human)**.

5 Experiments

In this section, we describe the experimental setup, datasets, metrics, and baselines results.

5.1 Setup

Environment. Experiments use Python 3.10, Hugging Face Transformers and FAISS. We cache embeddings to avoid redundant vectorization and index rebuilds.

5.2 Evaluation Dimensions

Our span/quote-anchored verification is consistent with QA-based factuality metrics such as QAFACET, which emphasize checkable evidence over free-form rationales (Fabbri et al., 2022). More broadly, the faithfulness literature argues for evaluations that privilege verifiability and attribution, a stance we adopt through Support, Function Classification, and Post-filter Citation Recall@ k alongside Citation Recall@1 (Li et al., 2022). We organize metrics along dimensions tailored to function-aware citation:

Citation Identification. Does the system select the ground-truth paper as the cited source? We report *Citation Recall@1*: the fraction of queries for which the top-1 function-aware *winner* equals the gold paper ($k=1$).

Retrieval Coverage. Is the correct paper even available among the LLM-supported candidates? We report *Post-filter Citation Recall@k*: for each query q , let S_q be the LLM-supported pool after filtering ($k = |S_q|$). The metric is the fraction of queries where the gold paper appears in S_q .

Function Support. Is the predicted *support* label (supported/unsupported) correct for each target function? Treating this as a binary classification per function, we report *accuracy*, *precision*, *recall*, and *F1*, with confusion matrices.

Citation Function. Does the predicted set of functions exactly match the gold set for the query? We report multi-label *Function classification accuracy*.

5.3 Baselines

Standard RAG (abstract-only). Our baseline is plain RAG system without function awareness. The retriever returns top- k abstracts (default $k = 10$),

Table 2: Function-aware pipeline — Support metrics (n=400 function targets).

Model (short)	F1	Acc	Prec	Recall
GPT-OSS-120B	0.849	0.758	0.881	0.820
Llama-4-Scout-17B	0.860	0.773	0.881	0.841
DeepSeek-R1	0.803	0.698	0.879	0.739
Teuken-7B	0.508	0.433	0.914	0.351

Short name → full ID: GPT-OSS-120B=openai/gpt-oss-120b; Llama-4-Scout-17B=meta-llama/Llama-4-Scout-17B-16E-Instruct; DeepSeek-R1=deepseek-ai/DeepSeek-R1; Teuken-7B=openGPT-X/Teuken-7B-instruct-research-v0.4.

which are provided as context. The prompt asks for a single cited paper.

Evaluation scope. Because this baseline has no notion of citation functions, we evaluate only citation identification and retrieval coverage in Table 1: (i) **Citation Recall@1**—fraction of instances where the *single cited paper* matches the gold paper; (ii) **Citation Recall@10**—fraction where the gold paper appears in the retrieved top- k .

Result comparisons with baseline are discussed in Section 8.

Table 1: Baseline RAG (abstract-only) — evaluation results.

Metric	Value
Citation Recall@1	0.83
Citation Recall@10	0.915

6 Results

In this section, we present end-to-end results of our function-aware pipeline on the FACET evaluation sets and compare against a standard RAG baseline. We apply the evaluation dimensions and metrics defined in Section 5.2 (Citation Recall@1, Post-filter Citation Recall@k, Citation Function Accuracy, and Function-Support F1).

6.1 Main Results

We evaluate four LLM backbones on the same held-out test set **FACET(gold)**. Full results are in Table 2 and Table 3, with a summary in Figure 4. **Llama-4-Scout-17B** is the strongest overall, followed closely by **GPT-OSS-120B**; **DeepSeek-R1** trails and shows stability issues (46/400 empty generations), while **Teuken-7B** underperforms. For detailed discussion, refer Section 7.

Confusion matrix (primary model). Figure 3 reports the binary *Support* confusion matrix for

Table 3: Function-aware pipeline: retrieval & function classification metrics (same test set). Recall@1: top-1 correctness. Post-filter Recall@k: gold paper appears in the LLM-supported pool after filtering (k is that pool’s size per query) and Function classification accuracy.

Model (short)	Citation Recall@1	Post-filter Citation Recall@k	Function Classification Acc
GPT-OSS-120B	0.622	0.835	0.918
Llama-4-Scout-17B	0.646	0.862	0.938
DeepSeek-R1	0.496	0.625	0.800
Teuken-7B	0.290	0.350	0.335

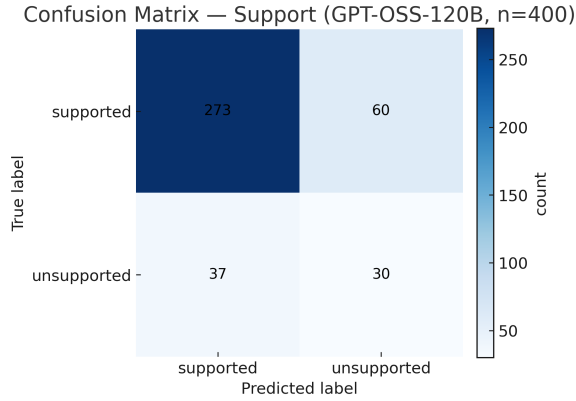


Figure 3: Binary *Support* confusion matrix for GPT-OSS-120B (n=400).

GPT-OSS-120B on the test set **FACET(gold)**. Cell counts are shown and blue intensity encodes magnitude. This view complements F1/Acc/Prec/Rec by showing where false positives vs. false negatives concentrate.

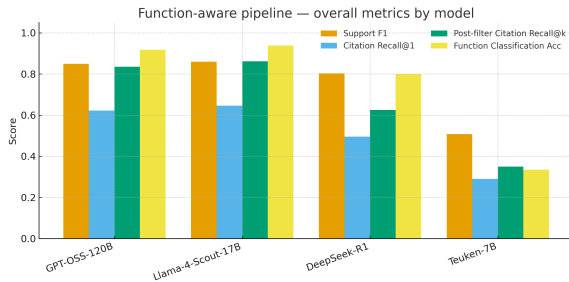


Figure 4: Overall comparison across models on the test set **FACET(gold)**. Bars show Support F1, Citation Recall@1, Post-filter Citation Recall@k, and Function Classification Accuracy.

6.2 Human-annotated subset

We further evaluate on a manually annotated sample: **FACET(Human)**. We reuse the same LLM-generated questions as in the main setup,

Table 4: Human-annotated subset — Support metrics for test set **FACET(Human)**.

Metric	F1	Acc	Prec	Rec
Value	0.815	0.710	0.842	0.790

Table 5: Human-annotated subset — identification & function classification.

Metric	Value
Function Classification Accuracy	0.990
Citation Recall@1	0.630

but the first author labels the function set and supported/unsupported from the abstracts. Scoring follows the metrics defined in Section 5.2.

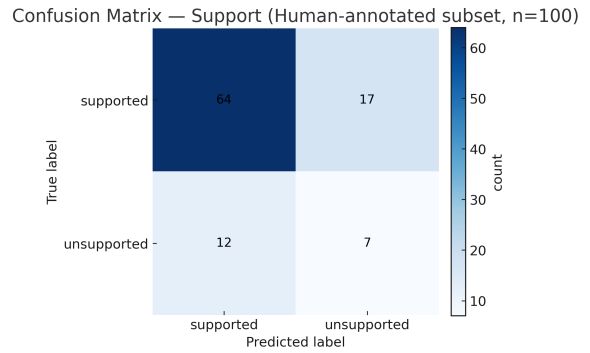


Figure 5: **Confusion matrix** (P=“supported”): TP=64, FP=12, TN=7, FN=17.

Interpretation. On the human-annotated subset, Support F1 (0.815) is consistent with main-test trends. *Function Classification Accuracy* (0.99) indicates close agreement with the human-written function sets, and *Citation Recall@1* (0.63) aligns with the stronger models on the full test set (Table 3), suggesting robustness against manually created ground truth.

7 Discussion

Tables 2–3 and Figure 4 indicate that the pipeline reliably identifies which function applies and whether evidence is present (e.g., *Function Classification Accuracy* ≈ 0.92 , *Support* precision ≈ 0.88 with F1 ≈ 0.85), but final top-1 paper choice remains weaker (*Citation Recall@1* = 0.622 vs. *Post-filter Citation Recall@k* = 0.835 for **GPT-OSS-120B**). A lower Citation Recall@1 does not imply the model’s chosen paper is incorrect. Many questions are answerable by multiple papers. When the model selects a non-gold paper that is still judged

supported by its rationale, the answer can be correct while the metric penalizes divergence from the single gold reference. In multi-source settings, *Post-filter Citation Recall@k* together with *Support* quality is therefore a better proxy for end-user utility than *Citation Recall@1* alone.

Ambiguity in COMPARES. Our dataset generally labels COMPARES only when an abstract makes an *external, cross-study* contrast (e.g., “compared to,” “unlike,” “in contrast with”). Models, however, often treat *internal, within-paper* contrasts as comparative. For example, in quant-ph/0102142, the sentence “the single quantum excitation case appears to be equivalent to a case with two discrete quasimodes” was flagged by the model as COMPARES, while our label was non-comparative. This difference in meaning (external vs. internal comparison) produces extra false positives and negatives for COMPARES and occasional swaps with EXTENDS/BACKGROUND. A potential solution could be to (i) report two COMPARES scores-*external* and *internal* and (ii) clarify annotation guidelines so internal regime/ablation contrasts are either counted under COMPARES (internal) or mapped consistently to BACKGROUND/EXTENDS.

Model variance and instability. Relative to *GPT-OSS-120B*, *Llama-4-Scout-17B* shows small but consistent gains (Tables 2, 3): Support F1 **+0.011** (0.860 vs. 0.849), Citation Recall@1 **+0.024** (0.646 vs. 0.622), Post-filter Citation Recall@k **+0.027** (0.862 vs. 0.835), and Function Classification Acc **+0.020** (0.938 vs. 0.918). A mild familiarity effect is possible because Llama-family prompts were used to author test queries, but the margin remains modest. **DeepSeek-R1** (F1 0.803, Recall@1 0.496, Post-filter Recall@k 0.625, Function Acc 0.800) also produced **46/400** empty answers despite retries, acting as hard false negatives. **Teuken-7B** is substantially weaker (e.g., F1 0.508; Recall@1 0.290).

Our primary test labels are LLM-derived under a strict protocol and post-processing. To probe robustness, we tested on human annotated test set FACET(Human). Its scores (Tables 4–5) mirror the main trends (e.g., Support F1 $\approx$ 0.815; Citation Recall@1 $\approx$ 0.63).

Observations on model size (Teuken-7B). **Teuken-7B** exhibits high precision but low recall on Support. This is consistent with (i) capacity limits of a 7B model in scientific abstract rea-

soning, leading to conservative positive decisions when cues are indirect, and (ii) domain mismatch (terminology-heavy physics/ML abstracts) that reduces sensitivity to weaker lexical signals. Net effect shows fewer false positives but many missed true supports, which in turn lowers *Citation Recall@1* even when candidates include valid answers.

Comparison to a standard RAG baseline. A plain RAG baseline (abstract-only; $k=10$ context) attains higher top-1 citation correctness (**Citation Recall@1** = 0.83) and strong retrieval coverage (**Retrieval Recall@10** = 0.915). Our function-aware pipeline, in contrast, enforces function-conditioned verification and thus reports lower top-1 selection for the primary model (**GPT-OSS-120B: Citation Recall@1** = 0.622) while still surfacing the gold paper in the supported pool with **Post-filter Citation Recall@k** = 0.835. This trade-off stems from (i) multiple valid sources per question and (ii) stricter evidence/provenance constraints tied to the predicted citation function. The baseline optimizes “what to cite” irrespective of rhetorical role; our method aligns “what” with “why,” improving interpretability at the cost of top-1. Closing this gap likely requires *function-aware retrieval*, not only function-aware filtering/generation.

8 Conclusion

We introduced a function-aware citation pipeline that predicts citation *functions*, retrieves candidates, verifies function-consistent evidence, and generates answers with function-tagged spans. On the main test set (FACET(gold)), **Llama-4-Scout-17B** performs best overall (Support F1 = 0.860, Citation Recall@1 = 0.646, Post-filter Citation Recall@k = 0.862, Function Classification Acc = 0.938), with **GPT-OSS-120B** close behind (0.849 / 0.622 / 0.835 / 0.918). A small human-annotated subset shows the same trend, indicating robustness. Compared to a standard RAG baseline (abstract-only) that achieves higher top-1 citation correctness, our pipeline *aligns citations to intent* via function-conditioned verification (see §7), and future gains will hinge on integrating *function-aware retrieval* to improve top-1 selection without sacrificing interpretability.

9 Limitations

Our retriever is not function-aware and relies on post-filtering, so function-relevant candidates

may be missed early. A potential solution could be a function-conditioned retriever (dual encoder with {query,function} inputs) or distillation from cross-encoder/re-ranker signals, including late-interaction models. The system uses title+abstract only, which can omit spans that justify a function. Moving to section-aware full-text retrieval with paragraph-level verification and span extraction would help. The gold evaluation set is modest and coarse (e.g., COMPARES conflates external vs. internal contrasts). Expanding to an expert-verified set with finer taxonomy, denser span labels, and broader domains would improve measurement. Our index covers 9,262 papers rather than the full unarXiv corpus (~2M), scaling to the full collection with function-aware metadata, section-aware chunking, and graph features (citation neighborhoods) would stress-test retrieval under shift. Metrics can be blunt in multi-source settings, where top-1 identification penalizes valid non-gold choices, so complementing Citation Recall@1 with Citation Recall@k and post-filter Citation Recall@k, plus per-function precision/recall/F1, cross-function confusion, and span-level attribution quality would yield a clearer picture. Finally, strict verbatim-quote rules improve auditability but reject paraphrastic evidence, and LLM variance introduces instability across backbones. Keeping strict mode for scoring while allowing calibrated relaxed matching (synonym/lemma spans), plus confidence modeling and light self-consistency checks, could reduce brittleness.

Acknowledgments

I would like to express my sincere gratitude to my supervisor, Tobias Schreieder, for his invaluable guidance, constructive feedback, and continuous support throughout this project. His insights and suggestions greatly improved both the quality of the research and the clarity of this paper.

GenAI Usage Disclosure

GPT-5 was employed strictly for linguistic assistance (grammar and spelling correction, minor phrasing, and document structure suggestions). The system did not generate ideas, design methods, conduct analyses, or shape the scientific claims. All text was subsequently verified and edited by the human authors, who bear full responsibility for the final publication.

Code and Data Availability

The code and evaluation dataset are available at [GitHub](#) (Joshi, 2025). Instructions for downloading and indexing the larger UnarXiv corpus (Saier and Färber, 2020) used in retrieval are also provided in the repository’s documentation.

References

- Samy Ateia and Udo Kruschwitz. 2024. [Bioragent: A retrieval-augmented generation system for showcasing generative query expansion and domain-specific search for scientific qa](#). *Preprint*, arXiv:2412.12358.
- Ines Besrour and collaborators. 2024. [unarxive_2024](#). https://huggingface.co/datasets/ines-besrour/unarxive_2024. Hugging Face dataset.
- Ines Besrour, Jingbo He, Tobias Schreieder, and Michael Faerber. 2025. [Squai: Scientific question-answering with multi-agent retrieval-augmented generation](#).
- Kritin Chatrinan, Thanapon Noraset, and Suppawong Tuarob. 2025. [Gan-cite: leveraging semi-supervised generative adversarial networks for citation function classification with limited data](#). *Scientometrics*, 130(2):679–703.
- Arman Cohan, Waleed Ammar, Madeleine van Zuylen, and Field Cady. 2019. [Structural scaffolds for citation intent classification in scientific publications](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 3586–3596, Minneapolis, Minnesota. Association for Computational Linguistics.
- Alexander Fabbri, Chien-Sheng Wu, Wenhao Liu, and Caiming Xiong. 2022. [QAFactEval: Improved QA-based factual consistency evaluation for summarization](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 2587–2601, Seattle, United States. Association for Computational Linguistics.
- Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. [Enabling large language models to generate text with citations](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6465–6488, Singapore. Association for Computational Linguistics.
- Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Meng Wang, and Haofen Wang. 2024. [Retrieval-augmented generation for large language models: A survey](#). *Preprint*, arXiv:2312.10997.

- Drahomira Herrmannova and et al. 2023. [unarxive 2022: All arxiv submissions and their rich metadata](#). In *Proceedings of the 2023 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*.
- Tomoki Ikoma and Shigeki Matsubara. 2023. [On the use of language models for function identification of citations in scholarly papers](#). In *Proceedings of the Second Workshop on Information Extraction from Scientific Publications*, pages 130–135, Bali, Indonesia. Association for Computational Linguistics.
- Gautier Izacard and Edouard Grave. 2021. [Leveraging passage retrieval with generative models for open domain question answering](#). *Preprint*, arXiv:2007.01282.
- Xiaorui Jiang, Chaoxiang Cai, Wenwen Fan, Tong Liu, and Jingqiang Chen. 2023. [Contextualised modelling for effective citation function classification](#). In *Proceedings of the 2022 6th International Conference on Natural Language Processing and Information Retrieval, NLPPIR '22*, page 93–103, New York, NY, USA. Association for Computing Machinery.
- Xiaorui Jiang and Jingqiang Chen. 2023. [Contextualised segment-wise citation function classification](#). *Scientometrics*, 128(9):5117–5158.
- Anushka Joshi. 2025. Function-aware citation retrieval and generation (code). <https://github.com/anushkajoshi909/function-aware-citation>. GitHub repository.
- David Jurgens, Srijan Kumar, Raine Hoover, Dan McFarland, and Dan Jurafsky. 2018. [Measuring the evolution of a scientific field through citation frames](#). *Transactions of the Association for Computational Linguistics*, 6:391–406.
- Vladimir Karpukhin, Barlas Ögüz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen tau Yih. 2020. [Dense passage retrieval for open-domain question answering](#). *Preprint*, arXiv:2004.04906.
- Paris Koloveas, Serafeim Chatzopoulos, Thanasis Vergoulis, and Christos Tryfonopoulos. 2025. [Can llms predict citation intent? an experimental analysis of in-context learning and fine-tuning on open llms](#). *Preprint*, arXiv:2502.14561.
- Avishek Lahiri, Debarshi Kumar Sanyal, and Imon Mukherjee. 2023. [Citeprompt: Using prompts to identify citation intent in scientific papers](#). *Preprint*, arXiv:2304.12730.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2021. [Retrieval-augmented generation for knowledge-intensive nlp tasks](#). *Preprint*, arXiv:2005.11401.
- Wei Li, Wenhao Wu, Moye Chen, Jiachen Liu, Xinyan Xiao, and Hua Wu. 2022. [Faithfulness in natural language generation: A systematic survey of analysis, evaluation and optimization methods](#). *Preprint*, arXiv:2203.05227.
- Jakub Lála, Odhran O’Donoghue, Aleksandar Shtedritski, Sam Cox, Samuel G. Rodrigues, and Andrew D. White. 2023. [Paperqa: Retrieval-augmented generative agent for scientific research](#). *Preprint*, arXiv:2312.07559.
- Vinzent Penzkofer and Timo Baumann. 2024. [Evaluating and fine-tuning retrieval-augmented language models to generate text with accurate citations](#). In *Proceedings of the 20th Conference on Natural Language Processing (KONVENS 2024)*, pages 57–64, Vienna, Austria. Association for Computational Linguistics.
- Haosheng Qian, Yixing Fan, Ruqing Zhang, and Jiafeng Guo. 2024. [On the capacity of citation generation by large language models](#). *Preprint*, arXiv:2410.11217.
- Ori Ram, Yoav Levine, Itay Dalmedigos, Dor Muhlgay, Amnon Shashua, Kevin Leyton-Brown, and Yoav Shoham. 2023. [In-context retrieval-augmented language models](#). *Preprint*, arXiv:2302.00083.
- Nils Reimers and Iryna Gurevych. 2019. [Sentence-bert: Sentence embeddings using siamese bert-networks](#). *Preprint*, arXiv:1908.10084.
- Tarek Saier and Michael Färber. 2020. [unarxive: a large scholarly data set with publications’ full-text, annotated in-text citations, and links to metadata](#). *Scientometrics*, 125(3):3085–3108.
- Zhihong Shao, Yeyun Gong, Yelong Shen, Minlie Huang, Nan Duan, and Weizhu Chen. 2023. [Enhancing retrieval-augmented large language models with iterative retrieval-generation synergy](#). In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 9248–9274, Singapore. Association for Computational Linguistics.
- Jiajun Shen, Tong Zhou, Yubo Chen, Delai Qiu, Shengping Liu, Kang Liu, and Jun Zhao. 2025. [Transpar-entize the internal and external knowledge utilization in LLMs with trustworthy citation](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 17858–17877, Vienna, Austria. Association for Computational Linguistics.
- Kurt Shuster, Spencer Poff, Moya Chen, Douwe Kiela, and Jason Weston. 2021. [Retrieval augmentation reduces hallucination in conversation](#). *Preprint*, arXiv:2104.07567.
- Simone Teufel, Advaith Siddharthan, and Dan Tidhar. 2006. [Automatic classification of citation function](#). In *Proceedings of the 2006 Conference on Empirical Methods in Natural Language Processing*, pages 103–110, Sydney, Australia. Association for Computational Linguistics.

- Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. [Beir: A heterogeneous benchmark for zero-shot evaluation of information retrieval models](#). *Preprint*, arXiv:2104.08663.
- David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. 2020. [Fact or fiction: Verifying scientific claims](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7534–7550, Online. Association for Computational Linguistics.
- Xiaoxuan Wang, Ziniu Hu, Pan Lu, Yanqiao Zhu, Jieyu Zhang, Satyen Subramaniam, Arjun R. Loomba, Shichang Zhang, Yizhou Sun, and Wei Wang. 2024. [Scibench: Evaluating college-level scientific problem-solving abilities of large language models](#). *Preprint*, arXiv:2307.10635.
- Xi Ye, Ruoxi Sun, Sercan Arik, and Tomas Pfister. 2024. [Effective large language model adaptation for improved grounding and citation generation](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6237–6251, Mexico City, Mexico. Association for Computational Linguistics.
- Yizhen Zheng, Huan Yee Koh, Jiaxin Ju, Anh T. N. Nguyen, Lauren T. May, Geoffrey I. Webb, and Shirui Pan. 2023. [Large language models for scientific synthesis, inference and explanation](#). *Preprint*, arXiv:2310.07984.

Appendix A: Implementation Tables

Table 6: SciBERT supervised baseline: model and training setup.

Item	Specification
Model	allenai/scibert_scivocab_uncased with BertForSequenceClassification (6-way)
Dataset	Citation-context data based on ACL-ARC subset with predefined train, val, and test files (each at 30%)
Label space	BACKGROUND, USES, COMPARES, MOTIVATION, CONTINUATION $\rightarrow$ EXTENDS, FUTURE $\rightarrow$ FUTUREWORK
Tokenization	Max length 256, padding to max length, truncation enabled
Optimizer	AdamW with learning rate 1×10^{-5} and $\epsilon = 10^{-8}$
Batch size	32
Epochs	13
Scheduler	Linear schedule without warmup
Loss	Cross-entropy
Sampling	<i>WeightedRandomSampler</i> with inverse-frequency weights for training and <i>SequentialSampler</i> for validation and test
Metric	Macro-F1 on the held-out test split
Result	Macro-F1 ≈ 0.86
Notes	Good context-level classification and easy to reproduce. We prefer the prompt-based LLM classifier for better downstream results and zero additional labeling.

Table 7: Label mapping used in experiments.

Source label	Mapped function in this paper
BACKGROUND	BACKGROUND
USES	USES
COMPARES	COMPARES
MOTIVATION	MOTIVATION
CONTINUATION	EXTENDS
FUTURE	FUTUREWORK

Stage / Role	Script	Input (key fields)	Output artifact
1) Function classification	classifying_question.py	query	classified_output.jsonl
2) Semantic retrieval	retrieval_query_based.py	query; FAISS index e5_index_subset_1/	topk_candidates_query.jsonl
3) Function-aware filtering	function_based_answer.py	top- k candidates; query	scored_candidates.jsonl
4) Answer synthesis	function_based_answer.py	supported candidates	final_answer.jsonl
Orchestration	run_pipeline_2.py	.env/config, CLI flags	pipeline_stdout.txt; artifacts/
Evaluation	eval_runner.py	final_answer.jsonl; gold set	eval_results.jsonl (reports Support P/R/F1, function_classification_accuracy, Citation recall@1, Post-filter Citation Recall@k)
Index build (offline)	build_index.py	papers.jsonl {paper_id, title, authors, year, abstract}	e5_index_subset_1/ (FAISS + metadata)

Table 8: Core components and intermediate artifacts of the function-aware citation pipeline. Interfaces are JSONL unless noted.